Surveying and Building an Automatic Knowledge Agent

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Table of Contents

1 Introduction..... 1

2 Evaluation of Asta Find Papers.....1

2.1 Methodology

2.2 Results

2.3 Observations and Comparison

3 Workflow of Asta Find Papers..... 3

3.1 Query Analysis

3.2 Execution

3.3 Paper Retrieval

3.4 Relevance Evaluation

3.5 Iterative Query Refinement

4 Implementing a Key Component..... 4

4.1 Description of the Task

4.2 State of the Art for the Task

4.3 Implementation of a Solution

4.4 Demonstration of Implemented Solution

4.5 Discussion and Ideas for Improvement

5 Conclusion.....5

References.....6

1 Introduction

There is a huge volume of academic literature but access to academic literature still remains a barrier in research despite the existence of services like Google Scholar. This is because Google Scholar takes experience and skill to be able to use effectively but even with that, it is hard to find suitable resources due to the vastness of academic literature. The development in terms of large language models has enabled new AI powered tools that can help automate this process and make academic papers more accessible. This report evaluates an AI powered knowledge agent, Asta, to investigate how it works and implement a key component in its workflow.

2 Evaluation of Asta Find Papers

2.1 Methodology

To evaluate Asta Find Papers' service, I compared it with Google Scholar by providing both services with the same two queries. The first query was *"feminist themes in Jane Austen novels"* that tested performance on humanities topics. The second query was "*Image Classification with MLP"* that tested Asta Find Papers' performance on looking up papers relevant to more technical queries. Both queries were run simultaneously on Google Scholar and Asta.

I compared both services based on the following criteria:

1. Response time and number of results
2. Presentation of results
3. Relevance and date of publication of the papers
4. Search transparency
5. Citation options

2.2 Results

Figure 1 shows the results generated by Asta Find Papers for query 1 (*"feminist themes in Jane Austen novels"*). Asta returned 37 papers in about 25 seconds. Similarly for the second query (*"Image Classification with MLP"*), Asta Find Papers returned 75 papers in 35 seconds. The results returned were categorized by relevance and showed tags for the exact topics each paper dealt with. There was also a section that showed evidence for the relevance tag of the paper, highlighting the most relevant portions from the text.

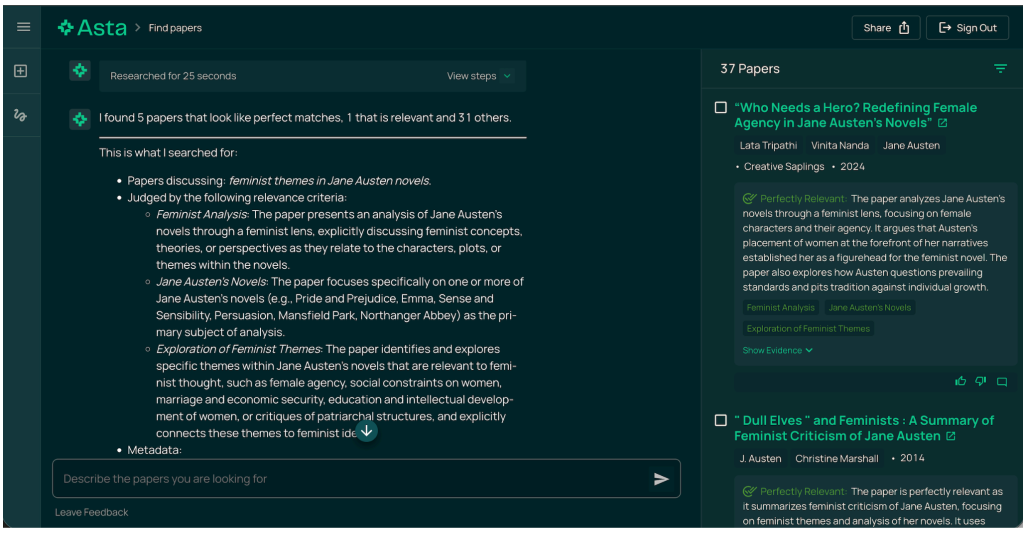

Figure 1. Results Generated by Asta for Query 1

Figure 2 shows the results generated by Google Scholar for the same query. Google Scholar returned about 32,000 results in 0.15 seconds. Google Scholar returned about 229,000 papers in 0.09 seconds for the second query. It highlighted the keywords that made the article relevant to the query. The cite option provided the same citation in multiple formats that could simply be copied. It also gave the option to view related articles and view different versions of the paper. There was also an option to save the paper.

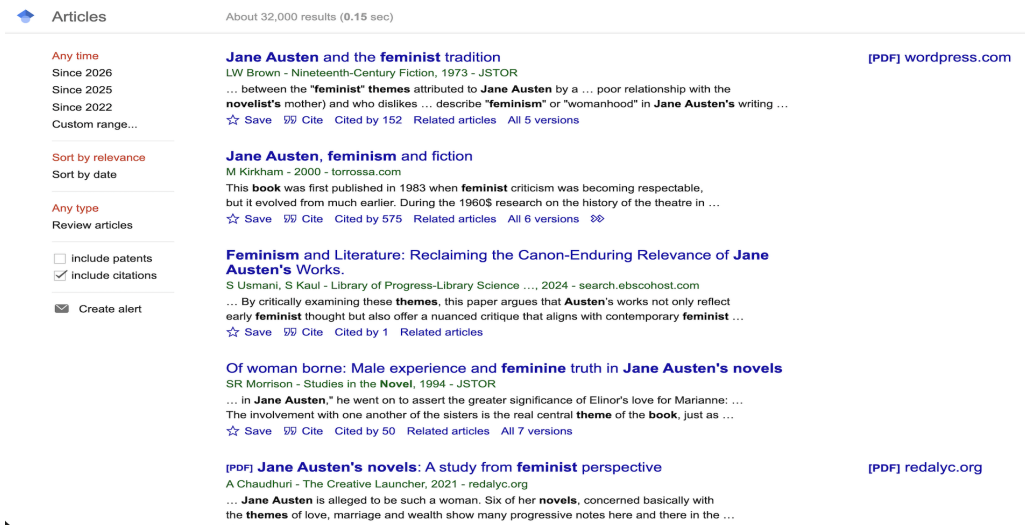

Figure 2. Results Generated by Google Scholar for Query 1

Table 1 shows a summary of the results obtained.

Table 1. Comparison between Asta Find Papers and Google Scholar

Criteria	**Asta Find Papers**	**Google Scholar**
Response Time (Query 1)	25 seconds	0.15 seconds
Response Time (Query 2)	35 seconds	0.09 seconds
Results (Query 1)	37	About 32,000
Results (Query 2)	75	About 229,000
Summary of the Paper	Yes	No
Search Transparency	Yes	No
Citation Formats	Only BibTeX	Multiple Options

2.3 Observations and Comparison

Google Scholar presented a much larger selection of papers in a shorter period of time compared to Asta as evident from *Table 1*. However, Asta was easier to navigate as it provided a tag for each paper that indicates how relevant each paper was to the current search query. Asta also provided evidence with relevant portions of the text to indicate why a particular paper got that particular tag. Moreover, Asta showed what it searched for and how it defined each concept while Google Scholar just presented the results. This made Asta's data retrieval process much more transparent.

Google Scholar had older publications that had more citations at the top of the list while Asta had a greater percentage of more recent papers. While Asta had a more intuitive user interface, it presented a smaller set of results which can make the user miss out on important papers that could have been useful. Some highly cited papers which came up in the Google Scholar search did not show up for the same Asta query.

3 Workflow of Asta Find Papers

Figure 3 illustrates the workflow of Asta Find Papers. Each step of the workflow is described in the subsequent sections.

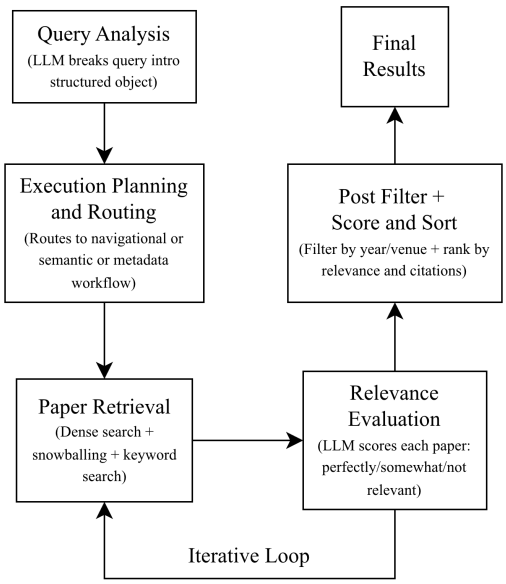

Figure 3. Workflow for Asta Find Papers

3.1 Query Analysis

Asta takes the raw query entered by the user and uses an LLM to break the query into a structured object. Then, according to Bragg et al. (2025), the query analyzer runs multiple short prompts in parallel with each targeting a specific subset of query properties. It extracts various properties such as whether the query is broad or navigational. It also extracts the semantic and relevance criteria along with the metadata constraints such as year or venue. Asta explicitly displays these criteria to the user as shown in *Figure 1*.

3.2 Execution

At this stage, the analyzed query is sent to an execution planner. The execution planner then routes the query to one of the several workflows (navigational, semantic, metadata, or author-based). Each of these workflows addresses a particular intent of the search. This particular feature of Asta makes it significantly different from Google Scholar which treats every query the same.

Each workflow then returns a set of candidates which are passed to the retrieval and relevance evaluation stages.

3.3 Paper Retrieval

The LLM generates k rephrasings of the user's query which is sent along with the original query to the semantic search API. Snippets from multiple queries are consolidated into a single paper item. Bragg et al. (2025) elaborates on how the snippets that reference other papers pull those cited papers also into the candidate set. Beyond this initial semantic search, Asta also uses other retrieval strategies like dense search, snowballing through forward and backward citation searches, keyword search and LLM suggested queries.

3.4 Relevance Evaluation

Asta uses a multi armed bandit approach to prioritize which candidate papers get evaluated first. Each paper is then judged individually by an LLM based on its title, abstract, and extracted snippets (Bragg et al., 2025). Papers are categorized into three tiers: perfectly relevant, somewhat relevant, and not relevant. The top results are then passed to the post filtering and ranking stage.

3.5 Iterative Query Refinement

Asta loops back and retrieves more papers if not enough highly relevant papers are found. This is done through citation tracking that involves snowballing forward and backward from already found relevant papers. Follow up queries are also generated based on the papers at the boundaries of the current search. This process stops when either enough highly relevant papers are found or when a predetermined limit is reached (Bragg et al., 2025).

4 Implementing a Key Component

4.1 Description of the Task

Query type classification is identifying whether a user's search query is navigational, semantic, or metadata based. This is an important step of the Asta workflow as it determines which specific workflow the query should be sent to. If the query is classified wrong, the paper finder will also return poor results. Natural language queries often are not very clear and modifiers like 'recent' or 'highly cited' can blur the boundaries between categories which makes this task challenging.

4.2 State of the Art for the Task

Query classification has been studied in a lot of search systems with traditional models like FastText being fast but not great with queries they haven't seen before. BERT-based models understand language better but are slower (Ning et al., 2023). More recently, systems like Asta Paper Finder use LLM based query analyzers to classify queries.

4.3 Implementation of a Solution

I used Gemini 3 Flash API to build an LLM based query classifier. It takes a query, sends it to Gemini with a prompt that defines the three categories (navigational, semantic, metadata) along with the requirements, and returns the classification.

The code along with the tests for the query classifier can be found in https://github.com/samridhiverma2504/Automatic-Knowledge-Agent.

The classifier was evaluated on 7 labeled queries across all three categories. Due to the API rate limit, I kept the testing dataset small but it was sufficient to demonstrate the effectiveness of the approach. The classifier achieved an accuracy of 6/7 (≈ 85.71%) on the test set.

4.4 Demonstration of Implemented Solution

Figure 4 shows the output returned by the classifier on a set of 7 queries.

```
Query: most cited papers on NLP from 2020
LLM Response: Metadata | Expected: Metadata

Query: the AlexNet paper
LLM Response: Navigational | Expected: Navigational

Query: MapReduce paper by Dean and Ghemawat
LLM Response: Navigational | Expected: Navigational

Query: image classification with MLP
LLM Response: Semantic | Expected: Semantic

Query: 2023 papers on query classification
LLM Response: Metadata | Expected: Metadata

Query: PyTorch optimizers
LLM Response: Semantic | Expected: Semantic

Query: recent papers on query classification
LLM Response: Semantic | Expected: Metadata

Accuracy: 85.71428571428571
```

Figure 4. Results Returned by the Query Classifier (with Category Examples)

4.5 Discussion and Ideas for Improvement

The LLM based approach worked well for queries that were clear. Ambiguous queries that used words like "recent" or "highly cited" caused misclassifications since the model sometimes interpreted these as metadata instead of semantic queries.

Initially, I provided the LLM with no examples for the 3 categories. This resulted in an accuracy of about 71.43% as shown in *Figure 5*. Adding two examples for each category improved accuracy to about 85.71% as shown in *Figure 4* which was an improvement by about 14%. This shows that adding more examples to the prompt would help improve accuracy further. Additionally, using a more diverse and larger test dataset would give a more accurate estimation of the accuracy of the query classifier. Another limitation of the current model is that it can't handle hybrid queries that blend multiple categories like searching for recent papers by a particular author on a particular topic which has both metadata and semantic components. Moreover, training a dedicated model like BERT on a dataset of query types would probably be faster and cheaper at a larger scale for a task like this.

```
Query: image classification with MLP
LLM Response: Semantic | Expected: Semantic

Query: most cited papers on NLP from 2020
LLM Response: Metadata | Expected: Metadata

Query: recent papers on query classification
LLM Response: Semantic | Expected: Metadata

Query: 2023 papers on query classification
LLM Response: Semantic | Expected: Metadata

Query: PyTorch optimizers
LLM Response: Semantic | Expected: Semantic

Query: MapReduce paper by Dean and Ghemawat
LLM Response: Navigational | Expected: Navigational

Query: the AlexNet paper
LLM Response: Navigational | Expected: Navigational

Accuracy: 71.42857142857143
```

Figure 5. Results Returned by the Query Classifier (without Category Examples)

Overall, this implementation does provide a strong baseline for query type classification that does not require any labeled training data.

5 Conclusion

This report evaluated Asta Find Papers and compared it to Google Scholar. It was found that while Google Scholar is faster and returns a greater volume of research, Asta provides a more curated selection of papers and is easier to navigate. On investigating Asta's workflow, I found that it involves query analysis, execution planning, paper retrieval, relevance evaluation, and iterative refinement. I identified query classification as a key component in this pipeline and wrote a Python script for an LLM based classifier that was implemented using the Gemini API. It achieved an accuracy of about 85.71% when the prompt included examples while it only achieved an accuracy of about 71.42% without examples. Future work could involve improving the prompt or training a dedicated model on a larger dataset of queries to improve speed and accuracy.

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